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Double descent, and statistical learning/physics

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Abstract

The analysis of learning action, machine learning and related practices in the field theoretically has been made by utilizing Computational Learning Theory (CoLT) Valiant 1984 and Statistical Learning Theory (SLT) Vapnik 1999. These two theories, while overlapped, provides a general framework in analysing and justifying learning actions and learning model constructions, aiding in the formation of modern practices. One of the famous insight using such framework is the *bias-variance tradeoff* Geman, Bienenstock, and Doursat 1992, which states that model complexity and generality inversely affect each other, thus guarantee the need for a safe bracket between them. However, recent literatures, Belkin et al. 2019a has indicated the fallout of such dilemma by the new phenomenon called *double descent*, where there exists an interpolation threshold that renders the current statistical justification for bias-variance inconsequential to a given region. Various 'anomaly' has also been detected similarly, with varying degree of sophistication and potential.In this paper, we would like to review through progress made in this particular field of research, and the phenomena potential explanations up until now.

Keywords: Statistical learning theory, double descent, deep learning, computational complexity theory, complexity theory, statistical physics, spin-glass models, theoretical physics

The modern machine learning landscape is dominated and guaranteed by the foundation of theoretical machine learning, specifically computational learning theory (CoLT) and statistical learning theory (SLT). The theory of machine learning and its modern practice has been developed and researched, of a substantial portion by empirical and heuristic approach, either by advancing new practices, architectures or by try-and-test modification. From its early onset of a regression estimator $\theta(x, y)$ on the linear regression problem, machine learning has developed substantially. Certain model concept with great successes includes regression-classification model, Bayesian modelling, generative learning model, support vector machine, Gaussian processes, and more. Of all such, the more formal and complex model architecture created, is the concept of a *neural network*. With increasingly sophisticated architecture, heuristic approach become popular, the fast-paced advancement of the field comes with new method, new results, new observations, and its far-reaching application which led to even bigger and larger scale deployment, there has been questions about the formation and status of a theoretical ground, a rigorous matter on the side of **theoretical machine learning**.

While rigorous and well-formulated in a sense, classical and theoretical machine learning was dwarfed by the modern advancement of machine learning as a whole, leading to several anecdotal problems regarding the interpretation of phenomena, the re-evaluation of the theory to fit the more updated analysis, and explanation to more sophisticatedly designed system. This and many more, plus as present, many of such advancements and improvements are heuristic, and the general theory and conceptual understanding remain limited, led to the choice to often opt for analogies and empirical workaround. Ultimately, this resulted in multiple 'failures' in explaining, interpreting, and predicting behaviours of new phenomena appeared in the modern landscape of machine learning. While there exists novelty of analysis, and different theoretical schemes to analyse such, particular problems remain in the form of irregularity and conflict of interpretations between frameworks.

We hope to use the theory of statistical physics, as seen of widespread usage in the history of development of artificial intelligence and computational method of such, as seen from Hopfield 1982; Amit, Gutfreund, and Sompolinsky 1985; Gardner and Derrida 1988; Seung, Sompolinsky, and Tishby 1992; Mezard, Parisi, and Virasoro 1987; Mezard and Montanari 2009; Engel and Van den Broeck 2001; Advani, Lahiri, and Ganguli 2013; Bahri et al. 2020; Mehta and Schwab 2014; Advani and Saxe 2017b; Li and Sompolinsky 2021; Ferraro et al. 2014, with mentions of such statistical mechanics application in Roberts, Yaida, and Hanin 2021. Some of those statistical learning schemes resulted in the works of G. E. Hinton on the variation of neural network called **Boltzmann machine**, as seen of Ackley, Hinton, and Sejnowski 1985 and Hinton 2012. Nevertheless, such question remains if the statistical scheme overreached of its current applicable form, to indict misleading or only limited improvement and interpretability onto a machine learning and in general an artificial intelligence system dynamics. In addition, assumptions and structural constraints, such as those incurred on Boltzmann machines for it to facilitate an environment interpretable of statistical physics remains a question of interest[a]. Even with such problems, we

a. It is widely concern of the fact that there exists many interpretations of machine learning mechanics under different formal language, or application-biased one. Such is, for example, information theoretic, as seen of Russo and would like to retain the view that statistical physics can be indeed used to interpret double descent, though with such phenomena, requires extreme cautions of logical assumptions and modelling setting.

1. Foundational works

On itself, the phenomena *double descent* has been investigated somewhat comprehensively, firstly introduced by Belkin et al. 2019a, which gives it distinctive saddle escape illustration. Further analysis was made by several literatures, particularly attributed the existence of double descent to the concept of model complexity and inductive bias, and potential explanations of such. Nakkiran et al. 2019 expanded the phenomena into deep neural network models. Their conclusion is reached by considering the perturbation of a learning procedure $\mathcal{T}$ on the effective model complexity $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ defined in the paper, separating the eventual phenomena into regions of observations, thereby in one way or another, predicting the tendency of double descent. This is also the first, arguably, "concise" definition of double descent, even though its nature is an empirical definition, including the notion of model complexity. Recently, there is also the Neural Tangent Kernel (Jacot, Gabriel, and Hongler 2018) which can be used to explain double descent, however further researches are still required. Preliminary works are done by Lafon and Thomas 2024, Schaeffer et al. 2023, and Liu and Flanigan 2023 on the role of optimization in double descent. Davies, Langosco, and Krueger 2023 attempted to unifying grokking with double descent, theorized a possibility of similarity during the generalization phase transition of the inference period, and Olmin and Lindsten 2024 attempted to explain epoch-wise double descent within the model of two-layer linear network. However, it is mentioned that it can also be expanded into deep nonlinear networks.

Further literatures on double descent varies, but fairly new, ever since their discovery articles in Belkin et al. 2019a and Nakkiran et al. 2019. Since then, we have Shi et al. 2024; Schaeffer et al. 2023; Quétu and Tartaglione 2023a; Lafon and Thomas 2024; Neal 2019; Davies, Langosco, and Krueger 2023; Quétu and Tartaglione 2023b; Liu and Flanigan 2023; Mei and Montanari 2020, 2019b; Adlam and Pennington 2020; Transtrum et al. 2025; Liu, Liao, and Suykens 2021; Allerbo 2025; Pezeshki et al. 2021; Brellmann et al. 2024; Nakkiran 2019a; Mei and Montanari 2019a; Heckel and Yilmaz 2020; Nakkiran et al. 2020; Belkin, Hsu, and Xu 2020; Yang and Suzuki 2024; Spiess, Imbens, and Venugopal 2023; Nakkiran et al. 2021; Y. S. Zhang 2024; Cherkassky and Lee 2024; Nakkiran 2019b. An even more exotic version of double descent, dubbed *triple descent* (or more), is discovered in Ascoli, Sagun, and Biroli 2020.

2. Foundation of learning theory

Most of the paper will argue about different features of statistical learning theory (Sterkenburg 2024; Vapnik 1999; Hajek and Raginsky 2021). It is then imperative to discuss the background setting of such. We are given the observations, or dataset of the form $\mathcal{S} = (\mathcal{X}, \mathcal{Y}) \subset \mathbb{R}^n \times \mathbb{R}^m$, of all 2-tuple pairs, assumed to be sampled or observed and governed by a distribution $\mathcal{D}$.

$$\mathcal{S} = \{(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)\} \subset \mathcal{X} \times \mathcal{Y}$$

The dataset is assumed to be i.i.d. sampling according to $\mathcal{D}$, which is unknown by the priori. Hence, we have the first set of assumption.

Assumption 2.1. *The observational space is governed by a particular sampling distribution $\mathcal{D}$, for fixed y-response according to the concept $c \in \mathcal{C}$. The sampling process is further assumed to hold no further structure, hence is identically and independently distributed (i.i.d.).*

The learning problem is then formulated as followed. Given the machine learning model expressed a hypothesis h of the hypothesis class $\mathcal{H}$, the learning theory aims for creating a procedure to learn either elements of the concept class $\mathcal{C}$ of all concepts $c : \mathcal{X} \to \mathcal{X}$, or the function class $\mathcal{F}$ of all functions $f : \mathcal{X} \to \mathcal{Y}$, which is usually set to $\{0, 1\}$ or $[0, 1]$. This distinction is trivial, hence, if it is clear, we will talk about the concept class $\mathcal{C}$ only as representative form. The learner $\mathcal{L}(h)$ consider the set of possible hypothesis $\mathcal{H}$, in which might not coincide with $\mathcal{C}$. It receives a partial image of sample $S = (x_1, \dots, x_2, \dots, x_n)$ drawn i.i.d. according to $\mathcal{D}$ as well as the label $(c(x_1), \dots, c(x_n))$. This constitutes the dataset $\mathcal{S}$, which are based on specific concept $c \in \mathcal{C}$ for the hypothesis to learn. The task is then to use (or *extract*) meaningful information to select a hypothesis $h_S \in \mathcal{H}$ that accurately mimic c, with marginal error Θ. The notation h_S stands for all hypothesis that can be inferred from the range of the dataset (the first argument, $\mathcal{X}$).

The marginal error Θ is considered of two parameters, the empirical error $\hat{R}(h)$ and the generalization error $R(h)$. We give the following definition of them.

Definition 2.1 (Empirical risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and a sample $S = (x_1, \dots, x_m)$. For some particular $\epsilon > 0$, the* ***empirical error*** *or empirical risk of h is defined*

Zou 2016; Xu and Raginsky 2017, categorical theoretic approaches, of Manin and Marcolli 2024, and accordingly, control theory, topological learning, graph theoretical, and statistical physics. While interesting as to illuminate the same problem under different lens, it is troubling of the messiness in such approach, as different theories claim different hypothesis, principles, theorems and axioms. Furthermore, it is of questionable intent to simply forgive the account of applying one's field on another — there exists the fundamental *domain bias* — of which as can be seen in statistical physics, for certain event to be applicable of using statistical law, one either must adapt the notion and introduce analogous notion (like temperature and relative model diffusing entropy), or restricting the domain, structure, range of expression, and mechanics — similar to how Boltzmann machines are. Additionally, applying without care gives increasingly large amount of domain bias, as concepts typically seen and assumed of a system in one theory might interject wrongfully, or simply conflict, narrow down the applicable range of such application, or in the worst case, invalidate the result.

by

$$\hat{R}_{\mathcal{S}}(h) = \underset{x\in\mathcal{S}\sim\mathcal{D}}{\mathbb{P}}[\ell\{h(x), c(x)\} \geq \epsilon] = \frac{1}{m}\sum_{i=1}^{m}\ell\{h(x_i), c(x_i)\} \quad (1)$$

The generalization risk is then defined as the extension of the empirical risk, of which instead of being evaluated on the set $\mathcal{S}$, it is evaluated on arbitrary infinite density sample space. The notation for the risk calculation can also be $R_{\mathcal{S}}$ instead, of which the need is to classify the set of which it is evaluated from.

Definition 2.2 (Generalization risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and an underlying distribution $\mathcal{D}$ on $\mathcal{X}$. For some particular $\epsilon > 0$, the generalization error or risk of h is defined by*

$$\begin{aligned} R(h) &= \underset{x\sim\mathcal{D}}{\mathbb{P}}[\ell\{h(x), c(x)\} \geq \epsilon] \\ &= \underset{x\sim\mathcal{D}}{\mathbb{E}}[\ell\{h(x), c(x)\}] \\ &= \int_{x\in\mathcal{D}} \ell\{h(x), c(x)\}\, dP(x) \end{aligned} \quad (2)$$

In essence, the empirical risk measure the hypothesis to observation error, and the generalization risk captures the difference of the hypothesis to the actual concept. Notice that by doing this, we have implicitly assumed that the observations and the concept is not the same thing, under the majority of scenarios. For the observations to be *approximately equal* or sufficiently captures the concept, there then would have to be various criteria to fulfil. For fixed $\mathcal{H}$, for fixed and sufficiently large $\mathcal{S}$, and no observation (data) errors, the empirical risk is the generalization risk. These two measures between h and c constitute the learning problem, which can also be separated into both cases - either empirical learning or generalization learning, one to minimize $\hat{R}(h)$, and one to minimize $R(h)$. Here, we emphasize such hierarchy by defining the generalization goal as the **nested criteria** for empirical risks minimization.

Definition 2.3 (Empirical learning problem). *We present the formal form of the empirical learning. Suppose we have a* ***target,*** *$c \in \mathcal{C}$, where $\mathcal{C}$ is an arbitrary concept class that captures targets of the same type. Suppose we are provided a set of observations $\mathcal{S}$. The problem is to use certain algorithm $\mathcal{A}$ using $\mathcal{S}$, to obtain a hypothesis h^* for a fixed $\mathcal{H}$ such that:*

$$R(h^*) = \min_{h\in\mathcal{H}} \hat{R}(h) = \min_{h\in\mathcal{H}} \underset{x\sim\mathcal{D}, x\in\mathcal{S}}{\mathbb{E}} \ell\{h(x), c(x)\} \quad (3)$$

The generalization best $\boldsymbol{h}$ is also called in literature as the *Bayes model*, such that it reduces the Bayes risk that is the infimum of the generalization risk, $R^* = \inf_{h\in\mathcal{H}} R(h)$. The hypothesis h^* is often called the *empirical best*, for it being the minimal, finite hypothesis of the lowest loss evaluation on the entire observation space $\mathcal{S}$. There exists no certified assumption regarding whether h^* aligns with the minimal generalization error.

Definition 2.4 (Generalization learning problem). *We present the formal form of the generalization learning problem. Suppose we have a* ***target,*** *$c \in \mathcal{C}$, where $\mathcal{C}$ is an arbitrary concept class that captures targets of the same type. Suppose we are provided a set of observations $\mathcal{S}$. Supposed we have an algorithm $\mathcal{A}$ that for fixed hypothesis space $\mathcal{H}$, equation 3 holds true. The problem is to use certain algorithm $\mathcal{A}'$ such that, under limited availability, to obtain* $\boldsymbol{h}$*, satisfies:*

$$R(\boldsymbol{h}) = \min_{h\in\mathcal{H}} R(\boldsymbol{h}) \leq \{\epsilon\}, \quad \epsilon > 0 \quad (4)$$

For a set of risk bounds ϵ. If the setting is deterministic, then there exists $\epsilon = 0$ accordingly.

Then, we can partially say that generalization problem is an advancement from empirical solution. As such, empirical solution imposes the observed concept, and not the concept itself. This is reflected in modern machine learning landscape, by modelling the concept as distributions, and then using notions such as KL-divergence to measure such relative disparity.

The process of generalization goal can be, described in certain form as the global optimization objective in which empiricism can obtain from reasonable set of assumptions and conditions. The following algorithmic framing indeed aims to focus and extradite such sense of operational goal.

Optimization problem 1: Empirical-generalization learning dynamics (non-proxy)

Data: $(x_i, y_i) \in \mathcal{S}, \Gamma \supset \mathcal{S}$, objective $\ell(y, f(x))$, $\nabla(\cdot)$ differentiable gradient field on ℓ.
Result: Expected optimized model to the minimal error on the objective $\ell(\cdot,\cdot)$, dependent on $\mathcal{S}$.
begin
  $\Gamma \supset \mathcal{S}$, the general set;
  $R(\cdot), \hat{R}(\cdot) \leftarrow \ell$ as objective functional;
  for *Fixed $\mathcal{H}$, with $h \in \mathcal{H}$, for $\epsilon > 0$* **do**
    Find $h \rightarrow \boldsymbol{h}'$, using algorithm $\mathcal{A}$ for minimization objective, such that

$$\mathcal{A}\left(\min_{h\in\mathcal{H}} \hat{R}_{\mathcal{S}}(h) + R(h)\right) \leq \epsilon \quad (5)$$

    for procedure $\mathcal{A}(\cdot,\cdot)$ of structural constraints.;
  end
end
return *Optimized model h on arbitrary objective space* (ℓ, ∇).

These two stages of learning procedure comes of as the intrinsic design of machine learning setting. As opposed to interpolation, where empirical learning problem would have to be put into first priority to achieve the best possible point-through model, machine learning leans on approximation more and a separated criterion that is only visible in the learning setting - the fact that we are using c' to extract c. This disparity is exactly the **generalization problem**, of which is then also the goal of machine learning, to approximate the general solution using the observed solution. One then can ask what is the difference between the two notions, and when we can guarantee that both will 'converge' to the same point. The process of **model selection** is then the procedure that will optimize $h \in \mathcal{H}$ to a given target fixture, such as the empirical best.

One of the major assumption or observation from the point of the hypothesis class, is that the hypothesis cannot know the generalization error. Thereby, the more efficient and often used procedure/algorithm in training, is then the *empirical risk minimization* (ERM) method, minimizing only the empirical risk to the empirical best; then, certain measures to 'generalize' this heuristically is apprehended on the hypothesis. The process of model selection is also the place that we get the concept of **bias-variance tradeoff**, and its longstanding form to be a foundational metric on model selection.

2.1 Empirical and structural algorithms

Suppose that for a concept class $\mathcal{H}$, we can then partition it to $\mathcal{H} = \bigcup_{k\in\mathbb{N}} \mathcal{H}_k$ for finite k hypotheses. Assume that they satisfy the uniform convergence (Shalev-Shwartz et al. 2010), which means in the arbitrary path landscape, one can guarantee ℓ-reduction, of m samples on $\mathcal{S}$ to:

$$\lim_{m\to\infty} \sup_{\mathcal{D}} \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m} \left[\sup_{h\in\mathcal{H}} \left[R(h) - R_S(h) \right] \right] = 0 \tag{6}$$

Then such is said to be possible as a ruling dynamic when for a learning problem, the empirical risks of hypotheses in the hypothesis class converges to their population risk uniformly, with a distribution-independent rate. The result of this is that a problem then can be considered learnable with the ERM rule, for any given characterization. We state the following definition.[b]

Definition 2.5. *A learning problem is learnable if there exist a learning rule $\mathcal{A}$ and a monotonically decreasing sequence $\epsilon_{cons}(m)$ such that $\epsilon_{cons}(m) \overset{m\to\infty}{\longrightarrow} 0$ and*

$$\forall \mathcal{D}, \quad \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m} \left[F(\mathcal{A}(S)) - F^* \right] \leq \epsilon_{cons}(m), \tag{7}$$

with $F^* = \inf_{h\in\mathcal{H}} R(h)$. A learning rule $\mathcal{A}$ for which this hold is denoted as a universally consistent learning rule.

There are certainly a fundamental class of rule or algorithm $\mathcal{A}$ to resolve this particular uniform convergence of classes. One of the major assumption or observation from the point of the hypothesis class, is that the hypothesis cannot know the generalization error. Thereby, one strategy is the *empirical risk minimization* (ERM) method, minimizing only the empirical risk to the empirical best; then, certain measures to 'generalize' this heuristically is apprehended on the hypothesis.

b. The reason why $r(\cdot)$ is not characterized in this preliminary section, is because of their uncertainty of effect. Hypothetically, $r(\cdot)$ is an additional structure (bias) of which apply a new gradient field in conjunction to the loss gradient field itself. This addition is uncertain of its content and analysis, since it depends on various stochastic dynamics of the model. We reserve from using regularization for now because of such reason.

Definition 2.6. *A rule $\mathcal{A}$ is an ERM (Empirical Risk Minimizer), denoted $\mathcal{A}_{ERM}$ if it minimizes the empirical risk $\hat{R}(h) = (1/m)\sum_{c\in C} \ell(h, c)$, over the observational space S,*

$$\hat{R}_S(\mathcal{A}_{ERM}(S)) = \hat{R}_S(h_S) = \inf_{h\in\mathcal{H}} \hat{R}_S(h) \tag{8}$$

Hence, the ERM tries to obtain $\mathrm{ERM}(h') = \arg\min_{h\in\mathcal{H}} \hat{R}(h)$.

The problem of overfitting still somewhat resides in the setting of the method. Hence, usually, we have an implicit term in addition, called the *regularizer*, of which is as followed.

Definition 2.7. *A rule $\mathcal{A}$ is an SRM (Structural Risk Minimizer), denoted $\mathcal{A}_{ERM}$ if it minimizes the empirical risk $\hat{R}(h) = (1/m)\sum_{c\in C} \ell(h, c)$, over the observational space S,*

$$\hat{R}_S(\mathcal{A}_{SRM}(S)) = \hat{R}_S(h_S) + \lambda r(h) = \inf_{h\in\mathcal{H}} \hat{R}_S(h) + \lambda r(h) \tag{9}$$

of the regularizing term $r(\cdot)$. Hence, the SRM tries to obtain $\mathrm{ERM}(h') = \arg\min_{h\in\mathcal{H}} \hat{R}(h)$, *while taking into consideration a forcing term r, usually defined on the model complexity.*

Hence, the SRM is inherently stronger than ERM. The regularizer $\lambda r(h)$, in specific case of convex loss landscape and reasonably well-behaved irregularities between domains, and of the structural regularization term $r_s(h) = \sum_{i,j} w_{ij}^2$ for weights of ij index, then we have that

$$\mathrm{ERM}\left(\inf_{h\in\mathcal{H}} \hat{R}_S(h), \ell\right) \approx o\left(\mathrm{SRM}\left(\inf_{h\in\mathcal{H}} \hat{R}_S(h) + \lambda r(h), \ell + \lambda r(h)\right)\right) \tag{10}$$

Where the little-o notation is used for saying of the optimization strength per arbitrary time-similar notion of algorithmic evolution. The reason for this change is because of the restriction domain that SRM indicted on the model optimization landscape. Effectively, we introduce the structural bias $\lambda r(h)$ that forces the morph of the total loss landscape. In well-behaved case, this is usually effective to a point, however in realistic system and pathological cases, such can lead to the c' approximation in c-space, just as we introduced in the reformulation of learning system. In this paper, we would like to presume the ERM method in question as the famous *gradient descent*. We refer to Achlioptas, n.d.; Ruder 2017 for general view of the algorithm, and J. Zhang 2019 for a source on particularly gradient descent algorithm on deep learning models.

2.2 Overfitting and underfitting

Following the above framework of statistical probabilistic learning theory, one understanding about the concept of underfitting and overfitting (similar to James et al. 2009) can be reached, using the notion of *partitioning* of observation space. Such work is typically done because of **practical condition**, in which the following constraints is applied.

Assumption 2.2. *No knowledge of $c \in C$ is provided.* [c]

c. Few points need to be said about this assumption. This assumption

Without such knowledge, $R(h)$ cannot be computed, accordingly. We specify such approach as *heuristic*, as it is only an approximation, and for better definition, we define it as followed.

Definition 2.8 (Heuristic empirical risk). *Given a dataset $\mathcal{S}$ of size m, supposedly of the concept $c \in \mathcal{C}$, and the hypothesis $h \in \mathcal{H}$, then the **heuristic empirical risk** take a partitioning $k < m$ with $k/m \geq 1/2$, hence defining the heuristic empirical loss on k, denoted $\hat{R}_k(h)$.*

Definition 2.9 (Heuristical generalization risk). *Given a dataset $\mathcal{S}$ of size m, supposedly of the concept $c \in \mathcal{C}$, and the hypothesis $h \in \mathcal{H}$. Fixed $\hat{R}_k(h)$, for the final optimization of algorithm $\mathcal{A}$, then the **heuristic generalization risk** take the remaining 2-partition of portion $p = k - m$, hence defining the heuristic generalization loss on p, denoted $\hat{R}_p(h_{\mathcal{A}})$.*

The two heuristic definition however, can be noted to not be equal to the true generalization risk - empirical risk pair in reality. Nevertheless, one can then define overfitting and underfitting in such manner.

Definition 2.10 (Underfitting). *A model h is called **underfitting** if for a constant $\epsilon > 0$ chosen for the setting, then $\hat{R}_k(h_{\mathcal{A}}) > \epsilon$, $\hat{R}_p(h_\alpha) > \epsilon$ over the entire observation space.*

Usually, underfitting is defined to be such that test error is smaller than training error partition, however, we do not think that is a good definition. Especially since, at least in this setting, the empirical error under algorithm $\mathcal{A}$ is reduced to the representative smallest error that $\mathcal{A}$ can achieve on that particular model, within particular setting of h, then come the heuristic generalization error. Considering also that $k/m \geq 1/2$, that is, at least the training partition is half of the observation set, then only under very specific case will $\hat{R}_p(h_\alpha)$ be smaller than $\hat{R}_k(h)$, statistically speaking. Thence this line of thought also reinforces the idea that both of them will be larger than the threshold ϵ. Usually, heuristically, this threshold of error is 0.3, normalized. Next, we define the notion of overfitting.

Definition 2.11 (Overfitting). *A model h is called **overfitting** if there exists a specific constant $\epsilon > 0$ chosen for the setting, such that $\hat{R}_k(h) < \epsilon$, $\hat{R}_p(h_\alpha) > \epsilon$ over the entire observation space.*

Namely, this mean that the static learning makes the generalization error worse. This draws the analogy of the student - studying explicitly for the past paper tests, yet unaware of the larger margin of questions in the large set of the test paper pool, leading to subpar performance. Similarly, overfitting is theorized to be fitting too fit to the reduced observation space - hence when it comes to the generalization observation space, it fails within such heuristic. The notion works on the mask of the risks, $\hat{R}_k$ and $\hat{R}_p(h_\alpha)$.

represents the *worst-case possible* of a system dynamic. From the perspective of a designer, inherent leakage of information can be provided in the design of a model. In such, case we are operating in the mode of a *resource-constrained, closed* game, in which the model with further assumptions of fixed class, operates on. The problem space itself is inherently large and penalize any additional structural changes. If one is to occur, we usually mention it as *implicit bias*, but another name that would be more fitting would be *structural attractor*, after the notion of gradient attractor in chaotic systems.

In practice, Assumption 2.2 means we cannot have $c \in \mathcal{C}$, or its probabilistic interpretation $\mathcal{D}$ such that $x \sim \mathcal{D}$, or in the extreme non-structured case, $(x, y) \sim \mathcal{D}$. In such case, the typical practical fix relies on the partitioning of the available partitioning space, that is, the empirical-generalization proxy. For large enough singleton count of $|\{(X_k, Y_k)\}|$, we assume that $\hat{R}_{\mathsf{Gen}}(h) \approx R(h)$. Thus, the algorithm can be stated in a form as Algorithm 2.

Optimization problem 2: Empirical-generalization learning dynamics (proxy)

Data: $(x_i, y_i) \in \mathcal{S}, y_i = \{\pm 1\}$, hinge objective $\ell(y, f(x))$, $\nabla(\cdot)$ differentiable gradient field on ℓ.
Result: Expected optimized model to the minimal error on the objective $\ell(\cdot, \cdot)$, dependent on $\mathcal{S}$.
begin
  $((X_1, Y_1), \dots)_n \leftarrow \mathsf{Partition}_n(\mathcal{S})$;
  $\{(X_j, Y_j)\} \leftarrow \mathsf{Train}$;
  $\{(X_k, Y_k)\} \leftarrow \mathsf{Gen}$ for $k \neq j$;
  for *Fixed $\mathcal{H}$, with $h \in \mathcal{H}$, for $\epsilon > 0$* **do**
    (Optimistically) achieve $h \to \boldsymbol{h}'$, using algorithm $\mathcal{A}$ for minimization objective, such that

$$\mathcal{A}\left(\min_{h \in \mathcal{H}} \hat{R}_{\mathsf{Train}}(h) + \hat{R}_{\mathsf{Gen}}(h)\right) \leq \epsilon \tag{11}$$

    for procedure $\mathcal{A}(\cdot, \cdot)$ of structural constraints.;
  end
end
return *Optimized model $h_{\mathcal{A}}$ on arbitrary objective space (ℓ, ∇).*

The assumption above of the generalization set to the generalization truth error is apparent in the usual system theorem, Theorem 2.3.

Theorem 2.3 (Mohri, Rostamizadeh, and Talwalkar 2012). *For fixed $\mathcal{H}$, for fixed and sufficiently large $\mathcal{S}$, and no observation errors, the empirical risk is the generalization risk:*

$$\mathop{\mathbb{E}}_{\mathcal{S} \sim \mathcal{D}^m} [\hat{R}_{\mathcal{S}}(h)] = R(h) \tag{12}$$

Proof. By linearity of the expectation, and the fact that we assume the sample is given i.i.d., we can write:

$$\begin{aligned} \mathop{\mathbb{E}}_{S \sim \mathcal{D}^m} [\hat{R}_S(h)] &= \frac{1}{m} \sum_{i=1}^{m} \mathop{\mathbb{E}}_{S \sim \mathcal{D}^m} \left[\mathbb{1}_{h(x_i) \neq c(x_i)}\right] \\ &= \frac{1}{m} \sum_{i=1}^{m} \mathop{\mathbb{E}}_{S \sim \mathcal{D}^m} \left[\mathbb{1}_{h(x) \neq c(x)}\right], \end{aligned} \tag{13}$$

for any x in sample S. Thus, we result in

$$\begin{aligned}\mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}[\hat{R}_S(h)] &= \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\left[\mathbb{1}_{h(x)\neq c(x)}\right] \\ &= \mathop{\mathbb{E}}_{x\sim\mathcal{D}}\left[\mathbb{1}_{h(x)\neq c(x)}\right] \\ &= R(h).\end{aligned} \tag{14}$$

which yields the desired result. □

This theorem can be transferred to a more generalized conjecture, of which fit the generalized notion of the empirical proxy and so on.

Conjecture 2.1. *Supposed $\mathcal{H}, \mathcal{C}$ are fixed, for $\mathcal{S}$ is the empirical c^* for $c \in \mathcal{C}$ the true concept. Then, for algorithm $\mathcal{A}(\cdot,\cdot)$ enforcing minimization apparatus, and the specific objective on the second argument, we have:*

$$\begin{aligned}&\mathop{\mathcal{A}}_{j\to\infty,\mathcal{S}_i\neq\mathcal{S}_k}\left(\min_{h\in\mathcal{H}}\hat{R}_{\mathsf{Train}}(h)+\hat{R}_{\mathsf{Gen}}(h),[\ell]\right)\\ &\approx \mathcal{A}\left(\min_{h\in\mathcal{H}}\hat{R}_{\mathcal{S}}(h)+R(h),[\ell]\right)+\epsilon_{\mathcal{S}}\end{aligned} \tag{15}$$

For $\mathcal{S}_i \neq \mathcal{S}_j$ such that there are only distinct samples, j the observation set size, $[\ell]$ the set of structural or empirical objectives enforced on the problem landscape, and $\epsilon_{\mathcal{S}}$ is the empirical minimum that the algorithm result can achieve assuming sufficient 'perfect interpolation'.

By the end of this section, we would also want to note that we deliberately choose an expression basis of $\mathcal{A}$ via the couple tuplet $(\min, [\ell])$ as a choice for *algorithm-agnostic class* of algorithm satisfies the dynamic only, not just implementations. The algorithm, in certain sense, depends much more on structural assumptions, of which is not accessible via observations but via design, and intrinsic information specified in the observation class itself. This is a requirement for fine-tuning analysis in the long-run, as there exists a myriad of different approaches.

3. Bias-variance tradeoff

In classical sources, James et al. 2009; Goodfellow et al. 2016; Hajek and Raginsky 2021; Mohri, Rostamizadeh, and Talwalkar 2012; Shalev-Shwartz and Ben-David 2014, the principle of bias-variance tradeoff is a classical principle used in the categorization of technique called *model selection*. Specifically, this refers to the progress of choosing the best predictor h^* that minimize the generalization error that is the main goal of a particular machine learning model, during the process of training and inference. Originally, the theory that bias and variance often come in to conjunction is **Estimation theory**, of which there exists an estimator $\hat{\theta}$ that use a set of observations, to estimate the concept, or the underlying mechanics of certain system that outputted the observation set, by the **probabilistic perspective** - that is, it is governed by appearance by an independently and identically distributed process of parameterized probability $\theta \in \Theta$. Here, parameterized means being expressed by a set, often finite, of parameters, by E. L. Lehmann 1998; Paninski 2005. It is only when Geman, Bienenstock, and Doursat 1992 introduced it to machine learning that we have an equivalent structure that is similar to bias-variance in machine learning.

3.1 Precursor (Geman, Bienenstock, and Doursat 1992)

The original definition of bias-variance tradeoff by Geman, Bienenstock, and Doursat 1992 is first constructed using the means-square error, which is regarded as a normal measure in the real encoding space. Their approach is to justify bias-variance via decomposition of the loss function ℓ, for such to find an alternative reasonable form of such loss landscape. Suppose of a regression problem to construct a hypothesis function $f(x)$ from $(x_1, y_1, \dots, x_N, y_N)$ for the purpose of generalization - that is, predicting unseen variational values for different pair $(x_j, ?)$ such that $? = y_j + \epsilon$ for a conceivable implicit error. To be explicit about the relation of this problem, or f on the given data $\mathcal{D} = \{(x_i, y_i) \mid i \leq N\}$, denote $f(x;\mathcal{D})$ instead of f, the natural mean-square measure as a predictor is:

$$\mathcal{M}(f,y) = \mathbb{E}\left[((y - f(x;\mathcal{D})))^2 \mid x, \mathcal{D}\right] \tag{16}$$

for $\mathbb{E}[\cdot]$ the expectation wrt to a distribution P. Decomposing the right-hand side, we have:

$$\begin{aligned}\mathcal{M}(f,y) &= \mathbb{E}\left[((y - f(x;\mathcal{D})))^2 \mid x, \mathcal{D}\right]\\ &= \mathbb{E}\left[(y - \mathbb{E}[y \mid x])^2 \mid x, \mathcal{D}\right] + (f(x;\mathcal{D}) - \mathbb{E}[y \mid x])^2\end{aligned} \tag{17}$$

Here, $\mathbb{E}\left[(y - \mathbb{E}[y \mid x])^2 \mid x, \mathcal{D}\right]$ does not depend on $\mathcal{D}$, but simply the statistical variance of y given x. The term $(f(x;\mathcal{D}) - \mathbb{E}[y \mid x])^2$ is considered a natural measure of effectiveness on $\mathbb{R}^n$ as a singular predictor of y. Now, for $\mathbb{E}_{\mathcal{D}}\left[(f(x;\mathcal{D}) - \mathbb{E}[y \mid x])^2\right]$ which depends on the training set $\mathcal{D}$ in its computation, is decomposed into the form of *bias-variance decomposition* terms, by derivation:

$$\begin{aligned}&\mathbb{E}_{\mathcal{D}}\left[(f(x;\mathcal{D}) - \mathbb{E}[y \mid x])^2\right]\\ &= \underbrace{\left\{\mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})] - \mathbb{E}[y \mid x]\right\}^2}_{\text{bias term}}\\ &+ \underbrace{\mathbb{E}_{\mathcal{D}}\left\{(f(x;\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})])^2\right\}}_{\text{variance term}}\end{aligned} \tag{18}$$

We summarize this in the following statement.

Theorem 3.1 (Bias-variance decomposition). *Suppose the model $f(x;\mathcal{D})$ for the data $\mathcal{D} = (x_i, y_i)$ and its parameter x is defined. For y_i of the target concept's responses y, and consider a regression problem with the loss measure $\mathcal{M}(f,y)$ of mean squared risk, the following statement is true:*

$$\begin{aligned}\mathbb{E}[\mathcal{M}(f,y)] &= \mathcal{B}(f,y) + \mathcal{V}(f,y)\\ &+ \mathbb{E}\Big[\mathbb{E}\left[((y - f(x;\mathcal{D})))^2 \mid x, \mathcal{D}\right]\Big]\end{aligned} \tag{19}$$

for $\mathbb{E}[\cdot \mid x, \mathcal{D}]$ *any expression with dependencies on* x *and* $\mathcal{D}$*. The bias and variance term is subsequently expressed by*

$$\mathcal{B}(f, y) = \underbrace{\{\mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})] - \mathbb{E}[y \mid x]\}^2}_{bias}, \tag{20}$$

$$\mathcal{V}(f, y) = \underbrace{\mathbb{E}_{\mathcal{D}}\left\{(f(x; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})])^2\right\}}_{variance} \tag{21}$$

The above decomposition principle is often expressed into a form where there exists the intrinsic noise Brown and Ali 2024:

$$\begin{aligned}\mathbb{E}_D\left[\mathbb{E}_{xy}\left(y - \hat{f}(x)\right)^2\right] &= \mathbb{E}_x\left[\left(y^* - \mathbb{E}_D[\hat{f}(x)]\right)^2\right] \\ &+ \mathbb{E}_x\left[\mathbb{E}_D\left(\hat{f}(x) - \mathbb{E}_D[\hat{f}(x)]\right)^2\right] \\ &+ \mathbb{E}_{xy}\left[(y - y^*)^2\right]\end{aligned} \tag{22}$$

In all, almost all representation of bias-variance decomposition are the same. For simplicity of notation, we adopt the similar form in the standard case of the paper Adlam and Pennington 2020:

$$\mathbb{E}\left[\hat{y}(\mathbf{x}) - y(\mathbf{x})\right]^2 = (\mathbb{E}\hat{y}(\mathbf{x}) - \mathbb{E}y(\mathbf{x}))^2 + \mathbb{V}\left[\hat{y}(\mathbf{x})\right] + \mathbb{V}[y(\mathbf{x})] \tag{23}$$

A main common theme of criticism toward bias-variance trade-off is the fact that the decomposition is much more general, and intrinsic for the class of *mean squared loss*. However, when considering the naturalness of an error measure and then its direct applicant, mean square error on real space of sufficient support and measure comes of as a very natural choice of an error measure, at least in consideration of the regression setting; for that, we then can also define classification as another top-layer above a regression's similar real continuous space, i.e. a decision layer. Furthermore, it can also be shown Brown and Ali 2024; Pfau 2013 that it also holds for the class of Bregman divergence measure.

What does the decomposition say about model selection principle in general? In general, bias-variance is typically presented of its decomposition only, and the asymptotic prediction of its behaviours. In fact, one of the reason that it became the rule-of-thumb for ML practitioner, as well as generally statistical learning (Lafon and Thomas 2024 provides a quite rigorous treatment of bias-variance tradeoff in the section on statistical learning theory) solidify the trade-off as a particular model selection principle. Generally, this tradeoff can be summarized as followed:

Theorem 3.2 (Bias-variance tradeoff)**.** *For the expected loss of any given hypothesis h, the bias* $\mathcal{B}(f, y)$ *and variance* $\mathcal{V}(f, y)$ *is inversely proportional, that is,* $\mathcal{B}(f, y) \propto \lambda^{-1}\mathcal{V}(f, y)$ *for some proportionality* λ *that may or may not be constant. In the most general case possible,* $\lambda = -1$ *on the entire error range.*

The tradeoff is then of inverse proportionality. Indeed, statistically, we have such tradeoff on a statistical framework in a more concrete sense. For the bias to increase, variance will increase, of which the criterion is inverse - we would like to have more bias but lower variance, and vice versa, according to such theory.

3.2 *Generalized decomposition*

According to P. Domingos 2000b, 2000a, the bias-variance decomposition can usually be treated as being controlled by three factor error, with their respective coefficients $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ for such:

$$\begin{aligned}&\mathbb{E}_{\mathcal{D}}\left[d((f, \mathbb{E}[y \mid x])\right] \\ &\quad= \lambda_1 \mathrm{Bias}(f, y) + \lambda_2 \mathrm{Var}(f, y) + \lambda_3 \epsilon(\mathcal{D}) \\ &\quad= \lambda_1 \underbrace{\{\mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})], \mathbb{E}[y \mid x]\}}_{\text{bias term}} \\ &\qquad+ \lambda_2 \underbrace{\mathbb{E}_{\mathcal{D}}\{(f(x; \mathcal{D}), \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})])\}}_{\text{variance term}} \\ &\qquad+ \underbrace{\lambda_3 \epsilon}_{\text{irreducible error}}\end{aligned}$$

where $\epsilon(\mathcal{D})$ is the irreducible error of the system, depends (theoretically) on the intrinsic imperfection of the dataset $\mathcal{D}$. This method of which the bias-variance-irreducible error are 'fit' into the total error by three coefficients $\lambda_1, \lambda_2, \lambda_3$ instead. This does not only scale up the B-V-IE triplet, but also leaves out the inexplainable gap between the scaling, and the actual normal measure.

4. Double descent

Nevertheless, of bias-variance and the analogous statistical learning theory concept, the target is the same. It is the dilemma of which is presented in **Occam's razor**, for choosing the sufficient model of good complexity, or bias, for tradeoff of its generalization ability, or variance. Then there must exist a sweet spot between the axis of bias and variance, since they are as exhibited above inversely proportional to each other. However, double descent seemingly broke the status quo, and insists on an interesting phenomenon - under the same setting, if we 'crank' the complexity high enough, we will then reach a point then called the **interpolation threshold**, such that the trend reverse and the error rate, instead of being theorized to go up, goes down to a certain line of lower bound.

The first identification of the double descent phenomena dated back to the paper of Belkin - Belkin et al. 2019a, in which the title is literally "reconciling" modern machine learning practice and the bias-variance tradeoff. In modern machine learning practice, or state-of-the-art developments, models are now bigger than ever. If to notice, we will see that currently models are inherently large, for example, a normal large language model will have from 900 millions (900M) to a few billions, for example 10 billions (10B) parameters. That is not taking into account the overall dynamics and structure of the model, which dictates the operating range and efficiency of the model itself. These model, based on the neural network architecture are somewhat trained to exactly fit (or interpolate) the data, almost certainly so that it turn from a prediction setting to an estimation setting. By statistical learning theory, this would be considered overfitting, and yet, they often obtain very high accuracy on test data.

The main finding that Belkin found is a pattern for how the apparent performance on unseen data depends on model capacity and the mechanism underlying the emergence of double descent. When function class capacity is below the "interpolation threshold", learned predictors exhibit the classical U-shaped curve from Figure 1. The 'modern' interpolating regime marks the opposite trend to the right, where the risk starts to decrease up to a lower bound, which then can be called the *optimal descent bound*.

> The bottom of the U is achieved at the sweet spot which balances the fit to the training data and the susceptibility to over-fitting: to the left of the sweet spot, predictors are under-fit, and immediately to the right, predictors are over-fit. When we increase the function class capacity high enough (e.g., by increasing the number of features or the size of the neural network architecture), the learned predictors achieve (near) perfect fits to the training data—i.e., interpolation. Although the learned predictors obtained at the interpolation threshold typically have high risk, we show that increasing the function class capacity beyond this point leads to decreasing risk, typically going below the risk achieved at the sweet spot in the "classical" regime. (Belkin et al. 2019a)

At this point, we would like to also clarify the term interpolation threshold in this case. Usually, under the pretext of overfitting and underfitting, interpolation threshold means the following:

Definition 4.1 (Interpolation threshold). *Assuming a typical supervised learning setting, for a hypothesis h within class structure $\mathcal{H}$, and a supposed concept c of the class $\mathcal{C}$. Suppose $[X, Y] \subset c \in \mathcal{C}$ being the dataset, or observable space of c on which h is tested on. Under the* ***empirical generalization learning*** *procedure, partitioning $[X, Y]$ into $[X, Y]_1$ and $[X, Y]_2$, for error evaluated on the first set, $\hat{R}_1(h)$ called training error, and the second set, $\hat{R}_2(h)$ called test error, with optimization applied on the first set, then evaluated statically on the second, the* ***interpolation threshold*** *with respect to a particular complexity measure of h, $\mathcal{M}(h)$, is the point $\mathcal{M}_0(h)$ where it is possible for $\hat{R}_2(h)/\hat{R}_1(h) \approx 1$, for $\hat{R}_1(h) \ll \epsilon$, $\hat{R}_2(h) \ll \epsilon$, with $\epsilon \geq 0$.*

This basically means that what is learnt, with respect to the model complexity that gauge *how much* and *what* the model can learn, perfectly fits the concept c itself, from the observation space on its own. We can say that, that specific sense, we gain the assumption that the observation space is *fully representative* of the concept class, and the concept class is then fully realized by the model. Such point where that is possible, or there then such behaviour is possible, is called the interpolation point. The definition relies on the statement 'where it is possible', hence it still, does not give us a particularly sound notion on to how and what constitute such possibility. The term empirical generalization learning comes from the fact that the procedure in practice of estimating and optimizing train-test-validation set in itself, is an imperfect empirical approximation of generalization testing. While the fact that we cannot gauge generalization error, in a purely blind black-box setting is settled (Mohri, Rostamizadeh, and Talwalkar 2012; Sugiyama 2015; Shalev-Shwartz and Ben-David 2014), the method is then to try to mimic such by using generalization-via-hidden observations. To facilitate this, deliberately, the empirical learning procedure is only dynamic during training session, and static throughout the whole testing and validation testing section, and further on.

By default, in general theory, such interpolating point is dangerous. Not only because it is inherently difficult to know exactly what the dataset constitute, whether it is actually representative of the concept that gives rise to it, or the structural information, noises, and interferences in between. It is hence impossible, to a certain point, to truly know the effectiveness of the model beside interpolating the current dataset which forms the observable details.

Another prominent discovery to look at is Nakkiran et al. 2019, on the double descent of deep learning models. The paper serves as the first observation upon which double descent is observed on modern, deep learning theoretic models, especially complex one like Transformer or ResNet networks.

They define *effective model complexity* of $\mathcal{T}$ (w.r.t. distribution $\mathcal{D}$) to be the maximum number of samples n on which $\mathcal{T}$ achieves on average ≈ 0 *training error*. This is an entirely empirical definition, similarly per definition as VC-dimension.

Definition 4.2 (Effective Model Complexity). *The* Effective Model Complexity *(EMC) of a training procedure $\mathcal{T}$, with respect to distribution $\mathcal{D}$ and parameter $\epsilon > 0$, is defined as:*

$$\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T}) := \max\left\{n \mid \mathbb{E}_{S\sim\mathcal{D}^n}[\mathrm{Error}_S(\mathcal{T}(S))] \leq \epsilon\right\}$$

where $\mathrm{Error}_S(M)$ *is the mean error of model M on train samples S.*

Using this definition, their main hypothesis can then be stated as the following three-fold regions:

Hypothesis 4.1 (Generalized Double Descent hypothesis, informal). *For any natural data distribution $\mathcal{D}$, neural-network-based training procedure $\mathcal{T}$, and small $\epsilon > 0$, if we consider the task of predicting labels based on n samples from $\mathcal{D}$ then:*

Under-parameterized regime. *If* $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ *is sufficiently smaller than n, any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.*
Over-parameterized regime. *If* $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ *is sufficiently larger than n, any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.*
Critically parameterized regime. *If* $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T}) \approx n$, *then a perturbation of $\mathcal{T}$ that increases its effective complexity might decrease* **or increase** *the test error.*

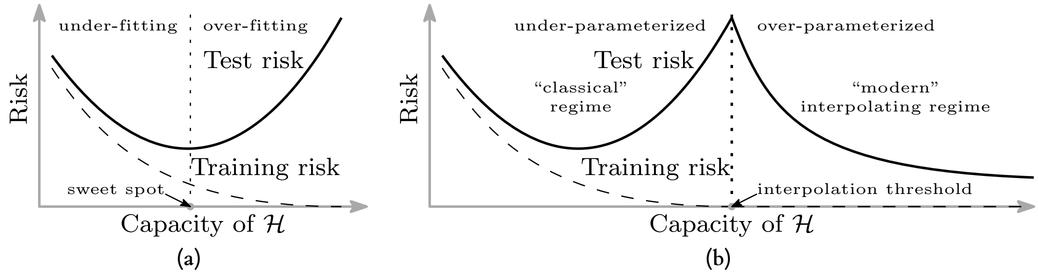

Figure 1. Curves for training risk (dashed line) and test risk (solid line). (**a**) The classical *U-shaped risk curve* arising from the bias-variance trade-off. (**b**) The *double descent risk curve*, which incorporates the U-shaped risk curve (i.e., the "classical" regime) together with the observed behaviour from using high capacity function classes (i.e., the "modern" interpolating regime), separated by the interpolation threshold. The predictors to the right of the interpolation threshold have zero training risk. Reproduced from Belkin et al. 2019a.

It is to notice that this definition is also only observational. By that, it only outlines the specific region of interest where the paradigm shift is identified through experimental results. It has no effective predictive power, or rather descriptive power more than setting up hypothesis with respect to the effective model complexity, and some arbitrary perturbation. Nakkiran et al. 2019 also noticed this difficulty in providing a theoretical definition and theorem regarding such hypothesis, as said in their manuscript. The existence of the critical regime is also not defined.

The behaviour itself has been particularly investigated, in certain settings. For example, before Belkin et al. 2019a, Advani and Saxe 2017a investigated the generalization error in neural networks of high-dimensional measure. Of such, various behaviours that bear similarity to double descent can be observed. Belkin, Ma, and Mandal 2018 expanded on his previous research on the particular subset of kernel learning models, providing a theoretical analysis of specifically the Laplacian kernel on standard neural network. Mei and Montanari 2020 investigated random feature regression in such regard, and also found similar results.

The fact that double descent is not well-defined itself is a problem on its own. Up to the author's knowledge and of myself, there has been no effective definition or given description that outlines specifically how double descent can be structured. Instability in reproducing experiments and the effective range of the phenomena remains a question on its own.

5. Statistical Physics: From Analogy to Isomorphism

In this section, we critically examine the intersection of statistical physics and machine learning. We posit that while statistical physics provides a rigorous framework for analyzing complex systems with many degrees of freedom, its application in machine learning has often been hindered by superficial analogies rather than mathematical isomorphisms. To truly leverage this framework, one must move beyond metaphorical comparisons and establish a dictionary of strict correspondence between physical quantities and learning dynamics.

5.1 *Reasons for Statistical Physics in Machine Learning*

The primary motivation for applying statistical physics to machine learning lies in the "curse of dimensionality" and the emergent phenomena observed in deep neural networks. Just as thermodynamics describes the macroscopic behavior of 10^{23} particles without tracking individual trajectories, statistical learning theory seeks to describe the generalization performance of models with billions of parameters. However, a critical mismatch often occurs in the abstraction level.

Common approaches often equate the "learning rate" in Stochastic Gradient Descent (SGD) to "temperature" in Langevin dynamics. We argue that this abstraction is insufficient and often misleading. In equilibrium physics, temperature is an isotropic scalar field. In contrast, the noise introduced by SGD is inherently anisotropic and state-dependent, scaled by the gradient covariance matrix. Therefore, a direct application of equilibrium concepts (like canonical distributions) without accounting for this structural mismatch leads to incorrect predictions about the system's stationary state.

To be more precise, let us consider the standard Langevin dynamics:

$$dW = -\nabla L(W)\, dt + \sqrt{2T}\, dB_t \tag{24}$$

where T is the temperature and dB_t is a Wiener process. The stationary distribution is the Boltzmann distribution $\rho(W) \propto \exp(-L(W)/T)$. However, SGD dynamics are fundamentally different:

$$W_{t+1} = W_t - \eta \nabla L_{\mathcal{B}}(W_t) \tag{25}$$

where $\mathcal{B}$ is a mini-batch. The noise covariance is:

$$\Sigma(W) = \mathbb{E}_{\mathcal{B}}\left[(\nabla L_{\mathcal{B}} - \nabla L)(\nabla L_{\mathcal{B}} - \nabla L)^{\top}\right] \tag{26}$$

This covariance is **state-dependent** and **anisotropic**, fundamentally different from isotropic thermal noise.

5.2 *Axioms and Formulation*

Statistical physics is built upon fundamental axioms, most notably the *Ergodic Hypothesis*, which assumes that a system evolves over time to explore all accessible microstates with a probability proportional to their Boltzmann weight.

The Mismatch. Deep learning dynamics, particularly under SGD, are fundamentally non-ergodic on practical timescales. The system does not explore the entire weight space but is confined to specific trajectories determined by the initialization and the landscape geometry.

The Isomorphism. To resolve this, we propose a strict isomorphism:

- **Microstate:** A specific configuration of weights $W \in \mathbb{R}^P$.
- **Hamiltonian (Energy):** The Loss function $L(W, \mathcal{D})$.
- **Quenched Disorder:** The dataset $\mathcal{D}$, which remains fixed during the training dynamics, analogous to the random interactions J_{ij} in spin-glass models.

This mapping situates machine learning not in the realm of simple ideal gases, but squarely within the physics of *disordered systems*, where the energy landscape is rugged, multi-valleyed, and dominated by metastability. The disorder average over datasets corresponds to the quenched average in spin-glass theory:

$$\overline{\langle O \rangle} = \int d\mathcal{D}\, P(\mathcal{D}) \langle O \rangle_{\mathcal{D}} \tag{27}$$

where $\langle \cdot \rangle_{\mathcal{D}}$ denotes the thermal average for fixed dataset $\mathcal{D}$.

5.3 Assumptions and Incompatibilities

Standard statistical mechanical treatments (e.g., the Replica Method) assume the system reaches thermodynamic equilibrium. This implies that the final solution is independent of the training path. However, the phenomenon of double descent—and specifically epoch-wise double descent—demonstrates that the "solution" is a transient state of a non-equilibrium process.

Assumption 5.1 (Equilibrium Assumption)**.** *The system reaches a stationary distribution $\rho^*(W)$ independent of initial conditions and training trajectory.*

This assumption is violated in practical deep learning for several reasons:

1. **Finite training time:** Training is stopped before equilibration.
2. **Non-convex landscape:** Multiple local minima prevent ergodic exploration.
3. **Time-dependent dynamics:** Learning rate schedules break time-translation invariance.

Forcing an equilibrium assumption onto a dynamic process like SGD is the fundamental error we aim to correct in subsequent sections.

6. Problem and Settings: Limitations of Equilibrium Theories

Here, we identify the specific failures of current theoretical frameworks when addressing the Double Descent (DD) phenomenon. While significant progress has been made, a comprehensive theory remains elusive due to a reliance on static analysis.

6.1 The Failure of Classical Bias-Variance

As established in Theorem 3.2, the classical U-shaped curve predicts that as model complexity increases beyond the optimal point, the variance term dominates and test error monotonically increases. Formally, for the decomposition:

$$\mathbb{E}[\mathcal{M}(f, y)] = \mathcal{B}(f, y) + \mathcal{V}(f, y) + \sigma^2_{\text{noise}} \tag{28}$$

classical theory predicts $\mathcal{V}(f, y) \to \infty$ as model complexity $\to \infty$. However, empirical evidence from Belkin et al. 2019a; Nakkiran et al. 2019 shows a second descent in test error for over-parameterized models ($P > N$).

6.2 The Limitation of RMT and Replica Theory

Recent applications of Random Matrix Theory (RMT) and Replica Theory have successfully derived the "peak" of the double descent curve at the interpolation threshold ($P \approx N$). For linear regression with random features, the expected test error takes the form:

$$R_{\text{test}} = \sigma^2 \frac{P}{N - P - 1} + \text{bias terms}, \quad P < N \tag{29}$$

which diverges as $P \to N$. However, these successes are limited to "static" or "shallow" models. They treat the solution as a global minimum (min-norm solution) found in a convex landscape:

$$W^* = \arg\min_{W} \|W\|^2 \quad \text{s.t.} \quad XW = Y \tag{30}$$

The Identification of Errors. The critical error in these approaches is the neglect of *feature learning dynamics*. Deep neural networks do not merely fit fixed features; they evolve their representation space. Furthermore, the existence of *epoch-wise double descent*—where the test error exhibits non-monotonic behavior over time even with fixed model size—proves that DD is a dynamical phenomenon. Equilibrium theories, by definition, cannot explain transient dynamical effects.

7. Identification: The Dynamic-Landscape Interaction

We categorize the Double Descent problem not merely as a statistical anomaly, but as a complex interaction between *multi-scale dynamics* and a *critical landscape*.

7.1 The Gap Between Statics and Dynamics

There exists a "gap" between the static description of the loss landscape and the dynamic trajectory of the optimizer. We identify three distinct regimes that must be unified:

Under-parameterized Regime ($P \ll N$): Dominated by simple signal learning. The landscape is relatively smooth with a unique global minimum.
Critical Regime ($P \approx N$) — Jamming: At $P \approx N$, the system undergoes a "Jamming Transition". The landscape becomes rugged with sharp minima, corresponding to satisfying

a precise number of constraints equal to the degrees of freedom. The condition number of the Hessian diverges:

$$\kappa(H) = \frac{\lambda_{\max}}{\lambda_{\min}} \to \infty \quad \text{as} \quad P \to N \tag{31}$$

Over-parameterized Regime ($P \gg N$): The system becomes "floppy" or unjammed. There exists a continuous manifold of zero-loss solutions, and the geometry of this manifold determines generalization.

7.2 Hypothesis: Double Descent as Phase Transition Navigation

Hypothesis 7.1 (Dynamic Origin of Double Descent). *Double Descent is the result of the SGD algorithm navigating through phase transitions in the loss landscape. Specifically:*

1. *The "peak" at $P \approx N$ is a trapping event in the jammed phase (sharp minima).*
2. *The second descent is a relaxation process driven by the specific noise structure of SGD, which allows the system to escape sharp minima and settle into flat, generalizing minima.*

The key insight is that SGD noise is not merely a nuisance but an *implicit regularizer* that performs geometric selection among the manifold of solutions.

8. Reformulation: Non-Equilibrium Statistical Physics Framework

To address the identified gaps, we reformulate the learning problem using a Non-Equilibrium Statistical Physics (NESP) framework. We move from describing static probability distributions to modeling the evolution of the probability density $\rho(W, t)$.

8.1 The Fokker-Planck Description

The evolution of the probability density over weight space is governed by the Fokker-Planck equation:

$$\frac{\partial \rho(W,t)}{\partial t} = \nabla \cdot (\rho \nabla L) + \nabla \cdot (D(W) \nabla \rho) \tag{32}$$

Crucially, the diffusion tensor $D(W)$ is not constant (as in thermal equilibrium) but is **state-dependent**, defined by the covariance of the stochastic gradients:

$$D(W) = \frac{\eta}{2B} \Sigma(W) \tag{33}$$

where η is the learning rate, B is the batch size, and $\Sigma(W)$ is the gradient noise covariance.

8.2 Noise-Driven Entropic Selection

This reformulation leads to a new mechanism for generalization. The SGD noise is anisotropic and approximately proportional to the Hessian (curvature) of the loss landscape:

$$\Sigma(W) \approx \frac{1}{N} \sum_{i=1}^{N} \nabla \ell_i \nabla \ell_i^\top \sim H(W) \quad \text{near minima} \tag{34}$$

This leads to curvature-dependent diffusion:

- **In Sharp Minima** (associated with the interpolation peak/jamming): The curvature is high, so $D(W)$ is large. This creates a strong effective "repulsive force" that drives the system out of these non-generalizing states.
- **In Flat Minima** (associated with the second descent): The curvature is low, so $D(W)$ is small. These states are dynamically stable under SGD.

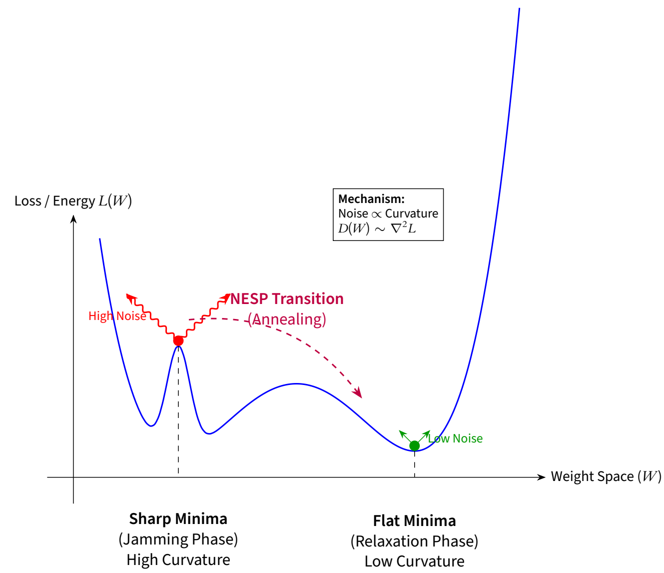

Figure 2. NESP Mechanism of Double Descent. At the interpolation threshold (Sharp Minima), high curvature induces strong anisotropic SGD noise (red wavy arrows), driving the system to escape. The system then relaxes to flat minima where low noise ensures stability. This process explains the second descent in test error.

8.3 The Mechanism of Double Descent

This framework reinterprets the Double Descent curve as a dynamical timeline:

1. **Descent 1 (Signal Learning):** Fast learning of dominant signal components (low-frequency modes). The model captures the primary structure in the data.
2. **Ascent (Peak/Jamming):** The system attempts to fit high-frequency noise but encounters the "Jamming transition" at the interpolation threshold. The landscape is rugged, and the trajectory is temporarily trapped in sharp, overfitting minima.
3. **Descent 2 (Relaxation):** In the over-parameterized regime, the "entropic selection" mechanism of SGD dominates. The landscape-dependent noise "anneals" the system, driving it out of sharp minima and into flat minima. This *implicit regularization* via non-equilibrium dynamics is the physical origin of the second descent.

Mathematically, the escape rate from a sharp minimum with curvature λ scales as:

$$\Gamma_{\text{escape}} \sim \exp\left(-\frac{\Delta L}{D_{\text{eff}}}\right) \sim \exp\left(-\frac{\Delta L \cdot B}{\eta \lambda}\right) \tag{35}$$

where ΔL is the barrier height. For sharp minima with large λ, the escape rate is high, explaining the instability of overfitting solutions.

8.4 Predictions and Implications

The NESP framework makes several testable predictions:

1. **Batch size dependence:** Larger batch sizes reduce noise ($D \propto 1/B$), making sharp minima more stable and potentially suppressing the second descent.
2. **Learning rate dependence:** Higher learning rates increase noise ($D \propto \eta$), accelerating escape from sharp minima and enhancing the second descent.
3. **Epoch-wise double descent:** The non-monotonic behavior over training time is a natural consequence of the system first falling into sharp minima (fast convergence) then slowly escaping (relaxation).

These predictions align with empirical observations in Nakkiran et al. 2019 and provide a unified explanation for both model-wise and epoch-wise double descent phenomena.

9. Experimental Verification: A Toy Model Proposal

The theoretical framework developed in preceding sections—positing that Double Descent arises from a Non-Equilibrium phase transition between Jammed and Relaxed states, mediated by anisotropic SGD noise—demands empirical validation. However, the complexity of modern deep neural networks obscures the fundamental mechanisms we seek to isolate. Therefore, we propose a series of *Gedankenexperimente* utilizing minimal, analytically tractable toy models that preserve the essential physics while eliminating confounding factors.

9.1 The Teacher-Student Protocol

We propose the **Teacher-Student framework** as our primary experimental testbed. This paradigm, extensively employed in the statistical physics of learning Engel and Van den Broeck 2001; Advani, Lahiri, and Ganguli 2013, provides the crucial advantage of a known ground truth, enabling precise measurement of generalization error without resort to heuristic train-test splits.

Setup. Consider a *Teacher network* $f^*(x) = \sigma(W^* x)$ with fixed, randomly drawn weights $W^* \in \mathbb{R}^{P^* \times d}$, where $\sigma(\cdot)$ denotes a nonlinearity (e.g., ReLU or tanh). The Teacher generates labels:

$$y = f^*(x) + \xi, \quad \xi \sim \mathcal{N}(0, \sigma_\xi^2) \tag{36}$$

where ξ represents irreducible label noise—a critical ingredient for inducing the interpolation peak.

The *Student network* $\hat{f}(x; W) = \sigma(Wx)$ with $W \in \mathbb{R}^{P \times d}$ is trained on N samples $\{(x_i, y_i)\}_{i=1}^N$ drawn i.i.d. from the Teacher's distribution. By varying the Student's capacity P while holding N fixed, we sweep across the interpolation threshold $P/N \approx 1$.

The Critical Control Parameter. The ratio $\alpha = P/N$ serves as our control parameter, analogous to the "packing fraction" in jamming physics. Our NESP framework predicts:

- $\alpha \ll 1$: Under-parameterized regime. Classical bias dominated behavior.
- $\alpha \approx 1$: Critical/Jammed regime. Interpolation threshold.
- $\alpha \gg 1$: Over-parameterized regime. Entropic selection dominates.

9.2 Physical Observables: Beyond Test Error

To validate the NESP mechanism—rather than merely confirming the existence of Double Descent—we must measure *physical observables* that directly probe the landscape geometry and noise structure. We propose three such observables:

9.2.1 Observable I: The Hessian Trace (Sharpness)

The trace of the Hessian, $\mathrm{Tr}(H) = \mathrm{Tr}(\nabla^2 L)$, provides a scalar summary of the local curvature at the solution W^*_{SGD}. This quantity is computationally accessible via Hutchinson's trace estimator:

$$\mathrm{Tr}(H) \approx \frac{1}{K} \sum_{k=1}^{K} v_k^\top H v_k, \quad v_k \sim \mathcal{N}(0, I) \tag{37}$$

where the Hessian-vector products Hv can be computed efficiently via automatic differentiation.

Hypothesis 9.1 (Sharpness Peak at Jamming)**.** *The Hessian trace, measured at the end of training, exhibits a pronounced maximum at the interpolation threshold:*

$$\mathrm{Tr}(H)|_{\alpha \approx 1} \gg \mathrm{Tr}(H)|_{\alpha \ll 1} \quad \textit{and} \quad \mathrm{Tr}(H)|_{\alpha \approx 1} \gg \mathrm{Tr}(H)|_{\alpha \gg 1} \tag{38}$$

Furthermore, in the over-parameterized regime, we predict a ***monotonic decay****:*

$$\frac{d}{d\alpha} \mathrm{Tr}(H) < 0 \quad \textit{for} \quad \alpha > 1 \tag{39}$$

This decay directly evidences the system's migration from sharp (Jammed) to flat (Relaxed) minima as capacity increases.

9.2.2 Observable II: Noise-Hessian Alignment

The central claim of our NESP framework is that SGD noise is *not* thermal (isotropic) but rather *geometric* (aligned with the Hessian). To test this, we propose measuring the alignment between the noise covariance matrix $\Sigma(W)$ and the Hessian $H(W)$:

$$\mathcal{A}(\Sigma, H) = \frac{\mathrm{Tr}(\Sigma H)}{\|\Sigma\|_F \|H\|_F} \tag{40}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. For thermal noise, $\Sigma = \sigma^2 I$ and thus $\mathcal{A} \propto \mathrm{Tr}(H)/\|H\|_F$, which is independent of the noise structure. For geometry-dependent noise, we expect strong positive correlation.

Hypothesis 9.2 (Anisotropic Noise Alignment)**.** *The alignment $\mathcal{A}(\Sigma, H)$ is significantly elevated compared to the null hypothesis of* *isotropic noise:*

$$\mathcal{A}(\Sigma, H) \gg \mathcal{A}(\sigma^2 I, H) \quad \textit{across all } \alpha \tag{41}$$

*Moreover, this alignment should **strengthen** near the interpolation threshold, where the curvature structure is most pronounced. This would provide direct evidence that SGD performs geometric, not thermal, exploration of the loss landscape.*

9.2.3 Observable III: Eigenspectrum Evolution

The full eigenspectrum of the Hessian, $\{\lambda_i\}_{i=1}^P$, encodes richer information than the trace alone. We propose tracking the spectral density $\rho(\lambda) = (1/P)\sum_i \delta(\lambda - \lambda_i)$ across the α-sweep.

Hypothesis 9.3 (Spectral Signature of Phase Transition). *At the Jamming transition ($\alpha \approx 1$), the spectral density exhibits:*

1. *A **hard edge** at $\lambda = 0$, corresponding to the vanishing of the null space as constraints saturate degrees of freedom.*
2. ***Outlier eigenvalues** at large λ, corresponding to sharp directions associated with fitting noise.*

In the Relaxed phase ($\alpha \gg 1$), the spectrum should exhibit:

1. *An extensive **null space** (bulk of eigenvalues at $\lambda \approx 0$), corresponding to flat directions in the solution manifold.*
2. *Suppression of outliers, indicating the system has escaped sharp directions.*

9.3 Experimental Protocol

To empirically validate the Entropic Selection mechanism, we propose the following experimental protocol:

Experimental Protocol 3: NESP Validation via Teacher-Student Model

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Data: Teacher weights W^*, noise level σ_ξ, sample sizes N, capacity range $P \in [P_{\min}, P_{\max}]$.
Result: Measurements of $R_{\text{test}}(\alpha)$, $\text{Tr}(H)(\alpha)$, $\mathcal{A}(\Sigma, H)(\alpha)$, $\rho(\lambda; \alpha)$.
begin
  for $P \in \{P_{\min}, \ldots, P_{\max}\}$ **do**
    Initialize Student network with random weights W_0;
    Train via SGD: $W_{t+1} = W_t - \eta \nabla L_{\mathcal{B}}(W_t)$;
    `; // At convergence:`
    Compute $R_{\text{test}}(W_T) = \mathbb{E}_x[(f^*(x) - \hat{f}(x; W_T))^2]$;
    Compute $\text{Tr}(H)$ via Hutchinson estimator;
    Estimate $\Sigma(W_T)$ from mini-batch gradient variance;
    Compute alignment $\mathcal{A}(\Sigma, H)$;
    (Optional) Compute top-k Hessian eigenvalues via Lanczos iteration;
  end
end
return *Observables as functions of* $\alpha = P/N$.

Predicted Signatures. If our NESP framework is correct, the experimental results should exhibit the following correlated signatures:

1. $R_{\text{test}}(\alpha)$ displays the characteristic Double Descent curve.
2. $\text{Tr}(H)(\alpha)$ peaks at $\alpha \approx 1$ and decays for $\alpha > 1$, *anti-correlated* with the second descent.
3. $\mathcal{A}(\Sigma, H)$ remains elevated throughout, peaking near $\alpha \approx 1$.
4. The spectral density transitions from hard-edge (Jammed) to bulk-zero (Relaxed) as α crosses unity.

The simultaneous observation of these four signatures would constitute strong evidence for the NESP interpretation of Double Descent as a dynamical phase transition driven by geometry-dependent noise.

9.4 Alternative Minimal Model: Two-Layer Linear Networks

For maximal analytical tractability, we also propose the **two-layer linear network**:

$$\hat{f}(x; U, V) = UV^\top x, \quad U \in \mathbb{R}^{d_{\text{out}} \times k},\ V \in \mathbb{R}^{d_{\text{in}} \times k} \tag{42}$$

Despite being linear in the input-output map (equivalent to $W = UV^\top$), this model exhibits *nonlinear dynamics* under gradient descent due to the factorization. Recent work Advani and Saxe 2017a; Saxe, McClelland, and Ganguli 2014 has shown that this model captures essential features of deep learning dynamics, including staged learning and critical slowdowns.

The advantage of this model is that the Hessian and noise covariance can be computed *analytically* in certain regimes, enabling precise theoretical predictions against which experimental measurements can be compared. The NESP predictions for this model are:

- The "effective temperature" $T_{\text{eff}} = \eta \cdot \text{Tr}(\Sigma)/P$ should exhibit non-monotonic behavior, peaking at the interpolation threshold.
- The condition number $\kappa(H) = \lambda_{\max}/\lambda_{\min}$ should diverge as $\alpha \to 1$, signaling the Jamming transition.

10. Conclusion and Future Horizons

10.1 Synthesis: From Equilibrium to Non-Equilibrium

In this work, we have undertaken a critical re-examination of the theoretical foundations underlying the Double Descent phenomenon. Our central thesis is that the apparent "anomaly" of Double Descent—its violation of the classical bias-variance tradeoff—is not a failure of statistical physics *per se*, but rather a failure of **equilibrium** statistical physics inappropriately applied to an inherently **non-equilibrium** dynamical process.

We have argued that the dominant theoretical approaches—Random Matrix Theory, Replica Methods, and classical bias-variance decomposition—share a common, often implicit, assumption: that the learning algorithm converges to a stationary state characterized by a Boltzmann-like distribution over weight space. This equilibrium assumption, while mathematically convenient, fundamentally mischaracterizes the nature of Stochastic Gradient Descent.

Our reformulation, grounded in Non-Equilibrium Statistical Physics (NESP), makes three key departures from the equilibrium paradigm:

1. **From static distributions to dynamical evolution:** We model the probability density $\rho(W, t)$ via the Fokker-Planck equation, treating the "solution" as a transient state rather than an equilibrium configuration.
2. **From isotropic to anisotropic noise:** We recognize that SGD noise is not thermal but geometric—its covariance $\Sigma(W)$ is state-dependent and aligned with the loss landscape curvature $H(W)$.
3. **From energy minimization to entropic selection:** We posit that generalization arises not from finding the global minimum, but from the *dynamical stability* of flat minima under geometry-dependent noise. Sharp minima, despite potentially lower loss, are rendered unstable by their own curvature.

10.2 Bridging the Gap: The Play and The Stage

A persistent tension in theoretical machine learning has been the disconnect between two perspectives:

- **The Stage (Landscape):** Static analysis of the loss surface—its minima, saddle points, and global structure.
- **The Play (Dynamics):** The trajectory of the optimizer through weight space—its convergence, oscillations, and implicit biases.

Equilibrium theories privilege the Stage: they characterize the set of possible solutions and assume the optimizer will find them. But they cannot explain *which* solution is selected, nor *how* the selection depends on hyperparameters like learning rate and batch size.

The NESP framework bridges this gap by coupling the Stage to the Play. The landscape geometry (Hessian) determines the noise structure (diffusion tensor), which in turn shapes the dynamical trajectory. Double Descent emerges naturally from this coupling: the interpolation peak corresponds to a *Jamming transition* where landscape constraints saturate available degrees of freedom, and the second descent corresponds to *entropic relaxation* as the system escapes sharp minima driven by its own noise.

This perspective offers a resolution to the longstanding puzzle: *Why do over-parameterized models generalize?* Our answer: not because they find better minima (they often do not), but because SGD dynamics are *geometrically biased* toward flat regions of the solution manifold. Generalization is an emergent property of non-equilibrium dynamics, not a static property of the loss landscape.

10.3 Future Horizons

The NESP framework, if validated, opens several exciting avenues for future research:

10.3.1 Grokking as Extreme Relaxation

The phenomenon of "Grokking"—wherein models exhibit delayed generalization long after achieving zero training error Power et al. 2022—presents a striking challenge to equilibrium theories. From the NESP perspective, we speculate that Grokking represents an *extreme* instance of the Jamming-Relaxation transition.

Specifically, certain data structures (e.g., modular arithmetic) may induce loss landscapes where the Jammed phase is *metastable* over extended timescales. The system achieves interpolation (zero training loss) but remains trapped in sharp, non-generalizing minima. Generalization occurs only after sufficient time for the noise-driven annealing process to escape these metastable states.

This interpretation predicts that Grokking should be:

- **Accelerated** by increasing learning rate or decreasing batch size (higher effective noise).
- **Suppressed** by explicit regularization that pre-selects flat minima.
- **Accompanied** by a monotonic decrease in Hessian trace during the "grokking" phase.

10.3.2 Transformers and Large Language Models

The training dynamics of Large Language Models (LLMs) represent perhaps the most consequential—and least understood—application of deep learning. The NESP framework suggests several speculative connections:

Temperature in Sampling vs. Training. LLMs employ a "temperature" parameter T during inference to control the sharpness of the output distribution. Intriguingly, our framework identifies an "effective temperature" $T_{\text{eff}} = \eta \cdot \text{Tr}(\Sigma)/(2B)$ that governs training dynamics. We pose the question: *Is there a principled relationship between these two temperatures?*

One speculative hypothesis: models trained with higher T_{eff} (larger learning rate, smaller batch) may naturally produce sharper internal representations, requiring lower inference temperature for coherent outputs. This would establish a non-trivial link between training dynamics and inference behavior.

Scaling Laws and Phase Transitions. Empirical "scaling laws" Kaplan et al. 2020 describe how LLM performance improves with model size, data, and compute. From the NESP perspective, these smooth power-law scalings may obscure underlying *phase transitions* in the training dynamics. As model capacity crosses critical thresholds relative to data complexity, the system may undergo Jamming-Relaxation transitions analogous to Double Descent.

Understanding these transitions could inform more efficient training schedules: rather than smooth scaling of learning rate, one might employ *punctuated* schedules that exploit phase boundaries—high noise to escape Jammed phases, low noise to stabilize in Relaxed phases.

10.3.3 Toward a Unified Theory of Implicit Regularization

The NESP framework provides a candidate mechanism for "implicit regularization"—the empirical observation that SGD finds generalizing solutions without explicit regularization terms. Our entropic selection mechanism offers a *dynamical* explanation: SGD implicitly regularizes by destabilizing sharp minima.

A unified theory would connect this mechanism to other forms of implicit bias:

- **Minimum-norm solutions** in linear regression.
- **Max-margin classifiers** in linearly separable problems.
- **Low-rank bias** in matrix factorization.

We conjecture that all these biases arise from the same underlying principle: the geometry-dependent noise of SGD preferentially stabilizes solutions with specific structural properties (low norm, large margin, low rank) because these properties correlate with flat loss landscape geometry.

10.4 Concluding Remarks

Double Descent is not merely a curiosity of over-parameterized models; it is a *window* into the non-equilibrium physics of learning. The classical bias-variance tradeoff, while a valuable heuristic in the under-parameterized regime, represents an equilibrium approximation that breaks down precisely where modern deep learning operates.

We have proposed that the correct theoretical framework is Non-Equilibrium Statistical Physics, wherein the learner is not a thermodynamic system seeking equilibrium, but a driven system whose steady state is determined by the interplay of deterministic forces (gradient descent) and state-dependent fluctuations (SGD noise). The Double Descent curve is the macroscopic signature of a microscopic phase transition from Jammed to Relaxed states, mediated by the entropic selection mechanism.

This perspective—if borne out by the experimental protocols we have proposed—would represent a significant shift in how we understand generalization in deep learning: not as a property of the solution, but as an emergent phenomenon of the dynamics that found it.

11. Organization (paper writing only)

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The following outlines the organizational structure in proposal specifically for the statistical physics paper. There are two parts, few claims that we posit in such discussion.

1. We posit that statistical physics is a quite formal and general method, yet unrefined, and often is applied incorrectly in the framework of perspective of machine learning. There are certain situations (or rather frequently) that statistical physics do describe a deeper interpretation of a dynamical system, but on the other hand, forces on machine learning the **physical perspective** and description, and thus enables false claims, restricted claims, setup that is impossible, of assumptions that never satisfy the theory itself, to force everything into a physical system instead of machine learning, then claiming the reverse is the truth.
2. We claim that to understand statistical physics interpretation to machine learning or system, we need more than apply superficially, that is, to simply do the one-one mapping of this framework's concept to another.
3. We will highlight such mismatch in the analysis, both in details of interpretation, and both formal way of mathematical expression, proofs, or so on, concretely.
4. We will propose a way to look at such, particularly through the problem of double descent, where other theories or statistical physics interpretation fails, and the analogous similarity to other previous theories and attempts historically.
5. We then bring up forth a new theory, or rather, a new abstract framework to explain it. Couple with that are some toy experiments, to explain of the system we want to highlight.

Thus, the following sectioning is adopted as the formal form for the document. This includes sections on the introduction of statistical theory, the argument and concurrent (present theory) analysis, then reformulation, then transition to a new theory with experiments.

1. Introduction to Statistical Physics: Simply the introduction of the discussion about the main position, as to what is statistical physics, and the morphism to machine learning, or the learning theory.
   (a) Reasons of statistical physics. This is to dissect *what*, and why statistical physics is even applied, and what does it mean or how to specifically connect to machine learning. This includes the analysis of different *abstraction* level that must be made to apply certain parameters or so on of ML, to fit statistical physics (for example, what is required to *simplify* or *add details* into it such that the notion of temperature is borderline acceptable in ML algorithms?).
   (b) Axioms of statistical physics. What are the axioms and formulation that statistical physics use, from assuming properties of multi-particle interpretation, the dynamic allowed, what kind of dynamics internally and externally, etc., such that statistical physics is expressed most generally? And what can be of the application and mismatch of between?
   (c) Assumptions of statistical physics. Explicit assumptions list, and incompatibility explanation, derivation, analysis.
   (d) Relevant concepts that it uses in formation, assumptions, already established notions.
2. Problem and settings. Here, it is simple as to *identify the errors*, speaking out loud of the problems, the problems of people solving the problems, the fallacy that make everything fails, what can be retained, what actually works, its limitation and so on.
3. Identification: Strictly identify the problem solutions and identification on "what exactly the problem can be categorized", proposing hypothesis, solutions, there of.
4. Reformulation: Reformulate it using a new theory.

All of this works in conjunction with double descent as a medium of illustration. Specifically, the *problem and setting* part would have to utilize in-context illustration, which then would be the double descent experimental cases. So is *identification* and reformulation. The **reformulation** part is however much more general, but use double descent as a toy problem for test bed.

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